DATASHEET | GUARANTEED TECHNICAL PARTICULARS

SmartLoop-4

4 Loop Analogue Addressable Fire Alarm Control Panel

Analogue addressable. The SmartLoop-4 watches up to **512 addressable devices on 4 loops**. Every detector reports its own analogue value to the panel, so the panel knows the state of each device - healthy, pre-alarm, alarm, dirty or missing - and shows the **exact device and room** in alarm. Designed to RDSO/SPN/217/2025 Version 3.1, automatic fire detection & alarm system for signalling & telecom installations.



Highlights

- **512 devices** 4 loops x 128 analogue addressable devices; 20 % spare for expansion.
- **Pinpoint location** loop, address, device type and 32-character location text.
- **Isolators built in** every detector and call point; Class A loops.
- **RDSO action levels** 4 levels, two-detector coincidence, suppression delay 0-10 min.
- **512 + 512 event log** fire and other events with date and time; PC download.
- **Networked** TCP/IP to Zonal HQ, RS-485 sub-panels, USB, optional GSM SMS.
- **Degrade mode** loop cards sound the alarm even if the CPU fails.
- **Railway ready** 0-49 deg C, 85 % RH, IEC 61000-4-4, electrified sections.

Technical data

SYSTEM	
Type	Microcontroller based analogue addressable fire alarm control panel, modular construction
Loops	4 loops (one plug-in loop card per loop)
Devices per loop	128 analogue addressable devices, addresses 001 to 128
System capacity	512 addressable devices; at least 20 % spare capacity for future expansion
Device identification	32-bit unique serial number in every device (printed on its barcode label)
Addressing	Auto-learn from the panel: the panel finds every new device on the loop and gives it an address or address entry at the panel from the barcode label - no DIP switch, no hand-held programmer
Alarm response	Alarm shown and outputs operated within 10 s of detection
Fire location	Loop, device address, device type and 32-character location text of every device in alarm
Two-detector coincidence	Programmable pairs of adjacent detectors (cross-addressing): one detector = panel alert, both = confirmed fire
Action levels	4 programmable action levels (RDSO/SPN/217 cl. 4.9.16)
Degrade mode	Each loop card sounds the alarm on its own if the main CPU fails
LOOP	
Wiring	2-wire, Class A (loop) or Class B (radial)

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Cable	Fire-survival circuit-integrity cable, 2 core x 1.5 sq.mm minimum
Loop length	Up to 1.5 km with 1.5 sq.mm cable (loop resistance up to 50 ohm)
Loop supervision	Open circuit, short circuit, device missing, wrong device type, duplicate address, earth fault
Isolation	Short-circuit isolator in every detector and call point
Interference	Built-in noise rejection on the loop: no separate EMI circuit needed
DISPLAY AND CONTROLS	
Display	7.0-inch colour TFT smart display, 800 x 480 pixels, 262K colours, brightness adjustable Capacitive touch screen with on-screen alphanumeric keyboard
Front keys	RESET, SILENCE, OK and four arrow keys
Access control	Key switch (access level 2) and passwords (access level 3)
Front panel functions	Buzzer silence, alarm silence, alarm activate (evacuate), lamp test, reset
Status LEDs	MAINS ON, SYSTEM ON, LOW BATTERY, STAND BY ON, FIRE, FAULT, SILENCE, CHARGING Per loop: ON, FIRE, OPEN, SHORT (16 LEDs)
Shown on display	Disabled, test, earth fault, primary supply fault, secondary supply (battery) fault
Panel buzzer	Piezo-electric; separate patterns for fire and fault
EVENT MEMORY AND COMMUNICATION	
Event log	Minimum 512 fire events plus 512 fault and other events, with date and time; kept without power
Real-time clock	Battery backed; synchronised over the network
PC connection	USB, RS-485 and Ethernet (TCP/IP) - download events, view site layout, configure
Networking	TCP/IP to the Zonal / Divisional headquarters; RS-485 / TCP/IP to sub-panels
GSM (optional)	ES-GSM100: SMS to 5 or more programmable mobile numbers within 30 s, with retry
OUTPUTS	
Sounder circuits	2 monitored sounder circuits, 24 V DC, 1 A each
Fire relay	Potential-free changeover contact, 1 A at 30 V DC
Fault relay	Potential-free changeover contact, 1 A at 30 V DC; fail-safe
Programmable relays	3 NO + 3 NC potential-free contacts on the panel, at least 500 mA each; more on the loop through ES-RM100 modules
Suppression release	Programmable delay 0 to 10 min in steps of 0.5 min, per room
Auxiliary supply	24 V DC, 0.5 A, protected
POWER SUPPLY	
Primary supply	110 V or 230 V AC, +10 % / -15 %, 50 Hz
System voltage	24 V DC
Secondary supply	2 x 12 V sealed maintenance-free batteries, charged by the panel
Battery backup	24 h standby followed by 15 min alarm at full load (NFPA 72 cl. 10.6.7.2.1.2)
Changeover	Automatic, within 10 s, no loss of function
Supervision	Mains failure, charger fault, battery low (below 21.6 V), battery missing Battery deep-discharge protection; reverse polarity protection
PHYSICAL AND ENVIRONMENT	
Enclosure	Wall mounted CRCA sheet steel, powder coated, lockable door; IP rating _____
Size (W x H x D) / weight	_____ mm / _____ kg (without batteries)
Operating temperature	0 to 49 deg C
Relative humidity	Up to 85 % (non-condensing)
EMC	IEC 61000-4-4: 2 kV on power ports, 1 kV on signal, data and control ports

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Railway suitability	AC / DC electrified and non-electrified sections; non air-conditioned rooms
Construction	Plug-in cards; FR-4 PCBs, conformal coated, legend printed
Materials	No radioactive material

Action levels (RDSO/SPN/217/2025 cl. 4.9.16)

Level	Started by	Panel action
Level 1	One detector of a coincidence pair, or aspirating detector stage 1	Panel buzzer and display; FIRE LED flashing
Level 2	Aspirating detector stage 2	Level 1 + hooter cum strobe in the Station Master room
Level 3	Manual call point, or aspirating detector stage 3	All hooters and strobes; power shut-down outputs (if programmed)
Level 4	Both detectors of a coincidence pair, aspirating detector stage 4, LHS / LHD module	All hooters and strobes within 10 s; suppression release after the set delay (0 to 10 min)

Standards and design basis

Reference	Title
RDSO/SPN/217/2025 Version 3.1	Specification for Automatic Fire Detection & Alarm System for Signalling & Telecom Installations
RDSO/SPN/144	Safety and reliability requirements of electronic signalling equipment (environmental, insulation and high-voltage tests)
IS 2189:2008	Selection, installation and maintenance of automatic fire detection and alarm system - code of practice
IS 11360	Smoke detectors for use in automatic electrical fire alarm system
IS 2175	Heat sensitive fire detectors for use in automatic fire alarm system
EN 54-2 / EN 54-4	Control and indicating equipment / power supply equipment
NFPA 72	National Fire Alarm and Signaling Code (response time, secondary supply)
IEC 61000-4-4	Electrical fast transient / burst immunity

Certification

Item	Certificate / report number
Control panel listing / approval (UL, FM, VdS, LPCB or ISO 17065 body)	_____
Detector sensitivity test certificate	_____
Environmental, insulation and HV tests to RDSO/SPN/144	_____
Manufacturer quality system (ISO 9001)	_____

Marking (cl. 10): manufacturer name and address, year of manufacture, serial number, specification number and the connection diagram inside the door. **Warranty:** 24 months from the date of supply. **Spares and support:** 10 years.

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Address: _____ Phone: _____
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